

DongGeon (Lloyd) Kwak

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EDUCATION

Seoul National University of Science and Technology Feb. 2022 – Present
B.E. in Computer Science and Engineering; Minor in Applied Artificial Intelligence Seoul, Korea

RESEARCH EXPERIENCE

Undergraduate Researcher, SeoulTech Jul. 2025 – Present
Advisor: Prof. Ji-Hyeong Han, Dept. of Computer Science and Engineering Seoul, Korea

PATENTS

Development Contributor, Korean Patent Application No. [10-2022-0142398](#) KIPRIS

WORK EXPERIENCE

Software Developer, CHASER Nov. 2022 – Jan. 2023

Software Developer, Laver Jun. 2022 – Oct. 2022

- Developed an object detection system, from model development through to integration.
- Built backend services and APIs.
- Developed an IFC-to-XML converter that transforms building information models into XML.

PROJECTS

MakeYourBrick – Single-image to LEGO (LDraw .ldr) model generator Python, FastAPI, Docker

- Built a local FastAPI web app that turns one photo into a brick model: SAM2 interactive masks, SAM 3D Objects reconstruction, then Studio-style color-aware brick tiling.
- Added in-browser GLB preview, automatic base-size selection (16/24/32/48 studs), shadow softening before color quantization, and a mesh-to-LDraw CLI; packaged the GPU runtime in Docker with pytest coverage.

Frequency-Aware Preprocessing for Diffusion Image Detection Python, Deep Learning

- Detected Midjourney, Stable Diffusion and DALL-E images on the 115,000-image DeepDetect-2025 dataset (70:15:15 stratified split); found that pixel statistics differ between real and generated images and built a frequency-aware preprocessing pipeline around it.
- Raised test accuracy from 72.6% to 96.3% (ResNet-50), 60.7% to 88.9% (EfficientNet-B4) and 78.6% to 92.4% (XceptionNet).

Logisim MLP – MNIST digit classifier as a digital circuit Python, Logisim Evolution

- Implemented a 256-128-10 MLP (about 34,000 weights, ReLU) for 16×16 binarized MNIST digits entirely in Logisim Evolution, with trained weights loaded into ROM and a 7-segment result display.

SKILLS

Programming Python, C++, Java

Tools PyTorch, TensorFlow, OpenCV, NumPy, scikit-learn, FastAPI, Docker, Git and GitHub, pytest

Language Korean (Native); English (TOEIC Speaking, Level AL)

Certification Craftsman Programming, HRD Korea (Nov. 2023)